ELECTRIC LOCOMOTIVE WAP7

WAP7 is a Broad Gauge locomotive developed by Chittaranjan Locomotive Works by modifying the gear ratio. Brake rigging & loco software of WAG9 Class locomotive and provision of hotel facility. It is capable for hauling trains at speeds upto 140 kmph and now largely used by Zonal Railways. All the locomotives are fitted with AAR tight lock couplers and chain link couplers. Banaras Locomotive Works (BLW) Varanasi started manufacturing WAP7 class locomotive first built locomotive is WAP7#30509 VARUN.

Electric Engine

- Electrical System - 25 KV, 50Hz. AC Overhead
- Transformer - 7775 KVA Oil Forced Air Forced, Primary: 25 KV, Secondary: Traction 1269 V, Aux. 1000V, Filter 1154 V, Hotel Lead 960 V
- Tractive Effort (Tonne) - Starting 32.88T, Continuous 23.13 T

Transmission

- Power - 6000 HP on wheels
- Traction Converter - 2x3000 KVA, O/P voltage: 3 phase 0-150 Hz., 2180V, water cooled (wear mixed with ethylene glycol)
- Gear Ratio 20:72

Brakes

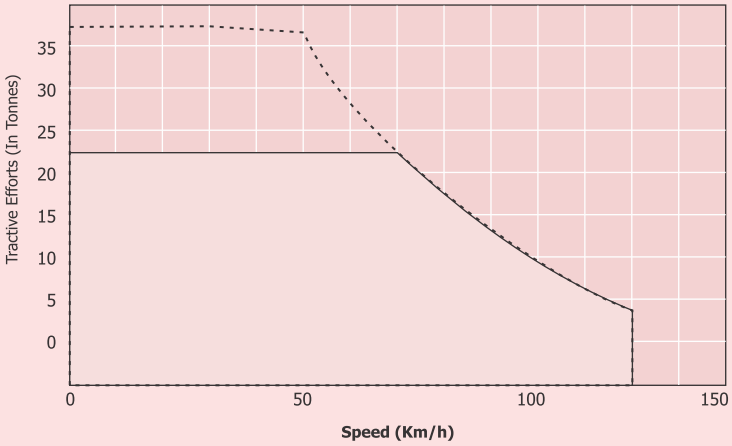
- Brake System - E - 70 (Faiveley make)
- Computer Controlled Brake (Knorr Bremse make)

Truck

- Bogies - CO-CO, Fabricated Flexi Coil, Bogie wheel base 3700 mm.

General Characteristics	
Electrical System	25 KV, 50Hz., AC overhead
Power	6000 HP on wheels
Tractive Effort (Tonne)	Starting 32.88 T, Continuous 23.13 T
Loco Weight	132 T (+/- 1%)
Speed	140 Kmph
Axle Load	20.5 Ton (+/-2%)
Length over headstocks	19280 mm
Body Width	3152 mm
Pantograph lock down height	4255 mm
Bogies	CO-CO, Fabricated Flexi Coil, Bogie wheel base 3700mm
Wheel diameter	1092mm new, 1016mm fully worm
Unsprung mass per axle	4.0 T
Gear Ratio	20:72
Brake System	E-70(Faiveley make) Computer Controlled Brake(Knorr Bremse make)
Transformer	7775 KVA Oil Forced Air Forced, Primary:25KV, Secodary: Traction 1269V, Aux. 1000V, Filter 1154V, Hotel Lead 960V
Traction Converter	2x3000 KVA, O/P Voltage: 3 Phase 0-150 Hz., 2180 V, Water Cooled (water mixed with ethylene glycol)
Hotel Converter	2x500 KVA, O/P Voltage: 750 +/- 6%, 50Hz. +/- 2%, THD<10%

Tractive Effort Chart



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